

A trustworthy longitudinal care record

One record; distinct role views; AI assists language, never clinical authority.

PROBLEM AND FIRST PRINCIPLES

From dated notes to a verifiable patient story

EHR snapshots and free-text consult notes fragment change over time. Nightingale organizes real-world Events, preserves parallel Artifacts, surfaces current priorities, and keeps every important derived statement traceable to raw source.

TIMELINE
What happened
Events ordered by when care happened;
Artifact activity stays inside the Event.

GLANCE
What matters now
Precomputed deterministic ranking with
reason, review state, and exact source.

PATIENT VIEW
What I need to do
Explicit patient-safe projection: instructions,
Tasks, follow-up, bounded Check-in.

TRUST-BOUND ARCHITECTURE

Patient

Staff

Clinician

Admin

FastAPI trust boundary
Session -> DB User -> authorize(action, clinic, patient)
Scope-first RBAC, exact-origin controls, request limits,
metadata-only audit, and explicit patient projections.

Longitudinal record

Patient -> Event -> Artifact -> exact Span

Comments | Versions | Audit | Tasks |
Highlights remain linked to the same Event
and source chain.

SINGLE AI EXIT
Redact -> LLMClient -> validate

- Names, IC/ID patterns, and phones are redacted before egress.
- Only mock and DeepSeek adapters; deterministic fallback is server-owned.

AUTHORITY BOUNDARY
Provider language, server decisions

- Provider may acknowledge and propose one bounded question.
- Server owns RBAC, safety, state, Task authority, limits, and provenance.

CORE JOURNEYS

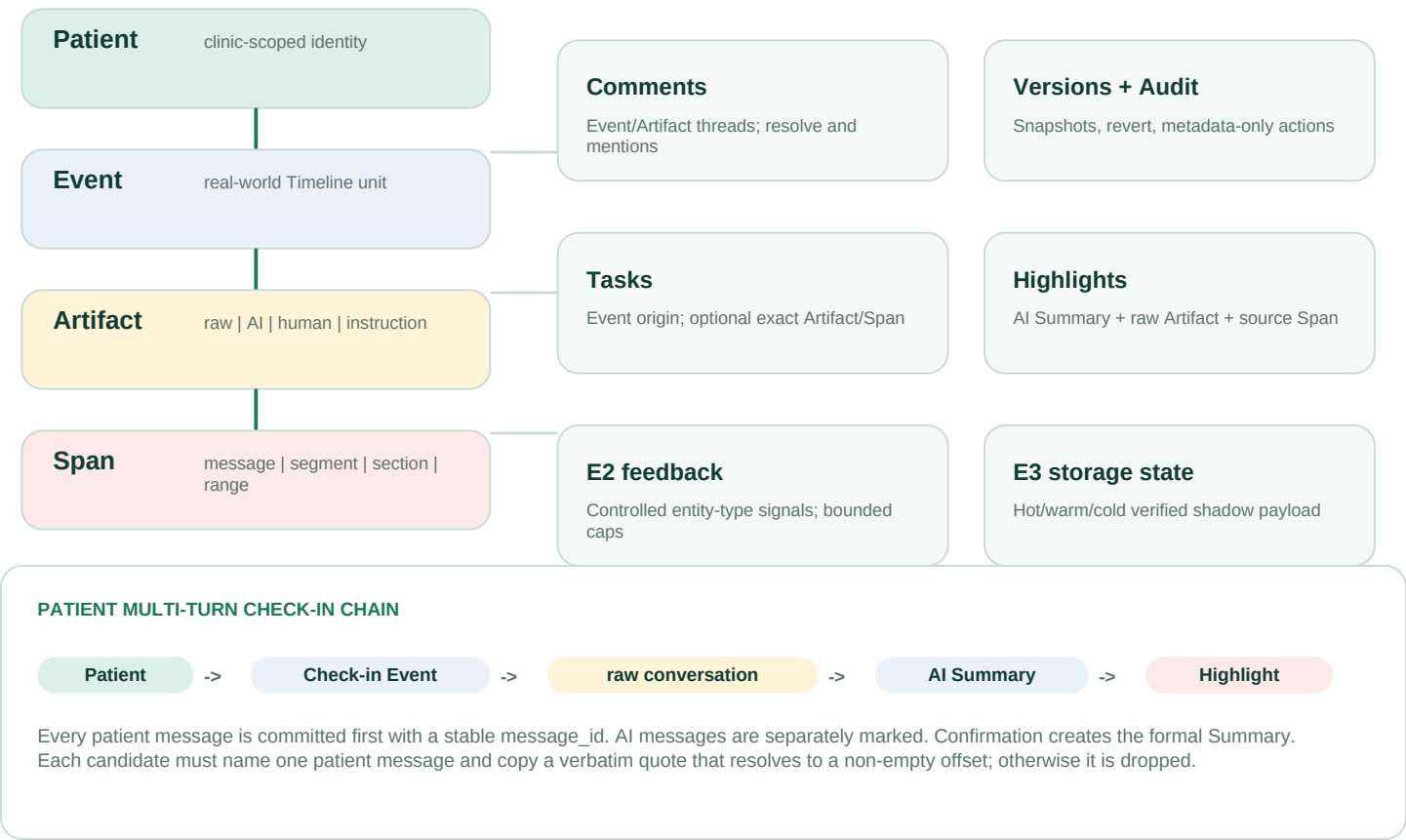
- Clinician: open Glance in under 10 seconds -> inspect exact source -> review Event artifacts and collaboration.
- Consult: normalize untrusted speaker-labelled text -> review -> confirm immutable Transcript -> AI Summary + exact candidates.
- Patient: understand instructions/Tasks -> complete a bounded raw-first Check-in -> clinician reviews originals, AI messages, Summary, and sources separately.

Prototype boundary
Synthetic data only. Not diagnosis, prescription, formal triage, clinic notification, production capacity, clinical validation, or regulatory compliance.

Data model, provenance, and bounded Check-in

Candidate Brief Entry maps to Event + parallel Artifacts; exact Span is the trust anchor.

COMPREHENSIVE SCHEMA



LIFECYCLE AND DETERMINISTIC AUTHORITY



At most four clarification questions. Patient may answer, supplement, correct, skip, end, return, confirm, or abandon. Diagnosis/medicine/dose/test requests use a fixed refusal. Transparent high-risk rules run without an LLM after raw persistence and stop ordinary questions.

RBAC AND AUTHORITY MATRIX

Role	Can	Cannot
Patient	Own safe view, visible Tasks, own Check-in	Internal notes/comments; other patient; clinical authoring
Staff	Same-clinic staff notes, Nurse Consult, Tasks	Clinician authorship; other clinic
Clinician	Same-clinic clinical review, notes, Copilot	Staff authorship; other clinic
Admin	Clinic identity/session/invite/access oversight	Clinical Note, Copilot, Task, or plan authoring

Clinician authority is permanent

AI/patient-derived information never overwrites clinician/staff Artifacts. Conflicts are flagged or clinician-authored content takes precedence.

What is verified, limited, or still missing

Observed local synthetic results are separated from mock, fallback, live Provider, and ASR non-runs.

CAPABILITY STATUS

E1 Role workspaces	IMPLEMENTED_AND_VERIFIED
E2 Importance weighting	IMPLEMENTED_AND_VERIFIED
E3 Data-decay shadow	IMPLEMENTED_AND_VERIFIED
E4 Local Voice	IMPLEMENTED_WITH_LIMITS
Patient Multi-turn Check-in	IMPLEMENTED_AND_VERIFIED
E5 Submission package	NOT COMPLETE - OWNER INPUT

CURRENT EXECUTABLE EVIDENCE

BACKEND 509 passed / 2 skipped Two skips require ignored local ASR model/audio inputs.	CHECK-IN 29 targeted tests Lifecycle, concurrency, raw-first, safety, RBAC, Provider, UI contracts.
SECURITY 22 security/integration Real session-cookie journeys and scope/log controls.	FRONTEND 3 Node checks / 59 modules Production TypeScript/Vite build passed.
GLANCE Layer B P95 5.115 ms 100 local samples after 10 warm-ups; target <= 300 ms.	STORAGE SQLCipher 4.12 / 18 tables Init, separate-key backup, rotated-key restore; plain reader blocked.

PROVIDER AND ASR EVIDENCE

mock + deterministic fallback - Full bounded Check-in journeys verified. - DeepSeek current turn succeeded live; strict final Summary failed and fell back. - D3 frozen Provider layer: NOT_RUN by design.	E4 Voice: implemented with limits - Default-off local faster-whisper; no diarization or invented confidence. - Historical synthetic local slice observed; current model/audio inputs absent. - Physical microphone and clinical/noisy accuracy: NOT RUN.
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SUBMISSION GATE Not Submission Ready	NEEDS_OWNER_INPUT: exact submitter name; actual 6-9 minute Demo Video file/link with full playback; final attachment/link access from the intended submitter context. No email, upload, sharing, or visibility change is authorized.
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